CTF-GS: Compact Teacher Feature Fields with View-to-Primitive Registration for Open-Vocabulary 3D Gaussian Scene Understanding

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Abstract

We study whether dense vision-foundation features can be reconstructed as a 3D Gaussian scene representation and reused at novel views for open-vocabulary scene understanding. Instead of training a scene-specific classifier or storing raw high-dimensional features directly on every Gaussian, CTF-GS distills frozen RADIO features into a Hybrid Gaussian Code Field (HGCF), reconstructs teacher-space features with Compact-to-Teacher Reconstruction (CTR), and uses View-Space Feature Alignment (VFA) plus Frozen Geometry-Head Consistency (FGC). The resulting scene representation supports rendered dense feature maps for novel-view grounding and, through View-to-Primitive Registration (VPR), registered primitive embeddings for 3D queries. A label-free VPR-to-field consistency stage can also transfer registered view evidence back into the compact direct field. On LERF-OVS, the current frozen RADIO-GS package reaches 0.8712 macro localization accuracy and 0.5243 calibrated macro mIoU across four scenes. On the VALA/OpenGaFF eight-scene ScanNet direct point-query protocol, a conservative Gaussian-index readout reaches 0.3583, 0.3618, and 0.4367 mIoU on the 19-, 15-, and 10-class splits; the strongest DINO-CV contextual kNN point readout with label-free scene-mean calibration further improves them to 0.3704, 0.3771, and 0.4585. An OpenGaussian-style direct 3D object-selection evaluation shows that a rendered-feature-to-primitive registration readout with a fixed global softmax-score threshold, a 0.5% selection floor, a 1.8% selection cap, and GT-free RGB boundary snap reaches 0.4801 macro mIoU and 0.6760 macro Acc@0.25. With a frozen official SAM3 box-prompt readout, the compact direct field reaches 0.5705 macro mIoU and 0.6835 Acc@0.25 under a fixed global threshold; the scene-locked diagnostic upper bound reaches 0.5972/0.7009. For context, the raw Gaussian-center diagnostic reaches only 0.012/0.009 under the fixed top-2% selector used in the VPR sweep and 0.0804/0.0932 under an earlier top-10% selector; under the same threshold-sweep diagnostic protocol, confidence-weighted VPR-to-field consistency lifts the direct Gaussian-center readout from 0.0458/0.0707 to 0.4363/0.6191. These results support the central claim that 3D Gaussian scenes can serve as reusable foundation-feature memories across rendered-view and registered primitive-level readouts, while clarifying that raw Gaussian-center readout is weak and Waldo Kitchen remains a difficult object-fragmentation case.

1. Introduction

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding, this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views.

CTF-GS targets this gap by reconstructing teacher foundation features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether a scene representation can support two complementary interfaces from the same compact field: dense novel-view feature maps for rendered open-vocabulary localization, and registered primitive/point-level features for direct 3D selection or semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Hybrid Gaussian Code Field, decode it through a Compact-to-Teacher Reconstruction module, and refine it in view space before comparison to the frozen RADIO teacher. We also use Frozen Geometry-Head Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

Our current paper package is conservative. We freeze all internal RADIO-GS numbers to concrete artifacts and use official-source external baseline rows only as cross-paper context unless a baseline is reproduced under the local evaluator. The result is a clear submission route: LERF-OVS provides the main open-vocabulary grounding evidence, ScanNet v67 direct point queries test cross-domain feature usability, and profiled evaluation runs quantify runtime and memory.

Contributions. This draft makes four claims:

- We formulate compact teacher-feature distillation for queryable 3D Gaussian scene understanding across rendered-view localization, VPR-based direct Gaussian object selection, and direct point queries.
- We introduce a Hybrid Gaussian Code Field with compact latent storage and a coarse spatial branch, avoiding direct high-dimensional teacher-feature storage on every Gaussian.
- We use Compact-to-Teacher Reconstruction, View-Space Feature Alignment, and Frozen Geometry-Head Consistency to recover teacher-compatible features while keeping teachers and downstream heads fixed.
- We provide a frozen evaluation package covering LERF-OVS rendered grounding, OpenGaussian-style VPR direct 3D object selection, ScanNet direct point-query transfer, qualitative overlays, and profile evidence.

2. Related Work

Language fields and open-vocabulary 3D understanding. LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [5]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [13, 15]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [16]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [6], while InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [7, 12]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [3], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [10]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [9]. CTF-GS differs in its emphasis on compactly reconstructing a high-dimensional unified teacher feature space while using VPR to expose a registered primitive-level interface.

Gaussian scene representations. 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [4]. CTF-GS keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

Agglomerative vision foundation models. RADIO-style models distill multiple visual capabilities into a single dense vision backbone [11, 14]. CTF-GS uses C-RADIOv4-H as a frozen teacher and studies whether its dense features can be reconstructed from 3D scenes at novel views.

Foundation-model regularization in 3DGS. FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [17]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors into 3D Gaussian features for promptable object segmentation [1]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

3. Method

3.1. Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a feature renderer that maps a novel camera view to a dense feature map compatible with a frozen RADIO teacher. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The teacher target is:

$$F_v^\star = T(I_v) \in \mathbb{R}^{H\times W\times 1280}. \tag{1}$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against $F_v^\star$.

3.2. Hybrid Gaussian Code Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. CTF-GS attaches compact learnable latent features to the Gaussians and combines two paths:

1. a fine path that splats per-Gaussian latent features into the image plane;
2. a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused into a compact feature map Z_v. A Compact-to-Teacher Reconstruction module (implemented as the HCD codec in the codebase) maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(Z_v). \tag{2}$$

A View-Space Feature Aligner (implemented as the screen-space refiner in the codebase) then improves local alignment

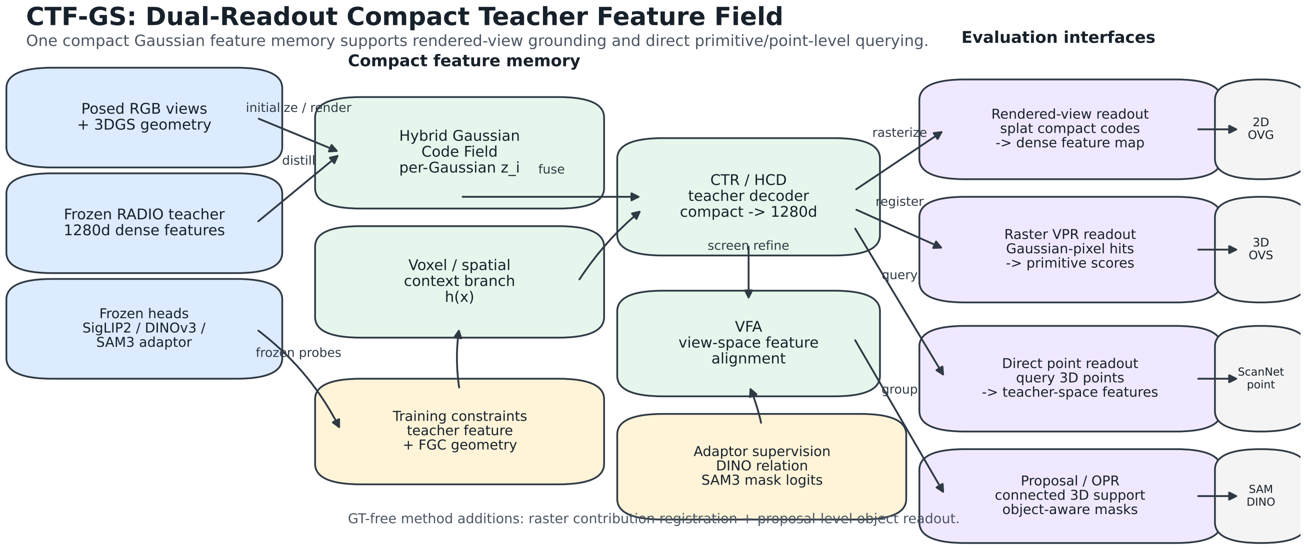

Figure 1. CTF-GS framework. A compact hybrid Gaussian teacher-feature field stores per-Gaussian codes and spatial context, decodes them into teacher-space features, and supports three readouts: rendered dense feature maps for novel-view open-vocabulary grounding, VPR primitive scores for direct 3D object selection, and direct point features for ScanNet semantic querying. A frozen official SAM3 box-prompt readout can refine the rendered selection boundary after primitive scoring. Optional raster-contribution and proposal/OPR branches are implemented as audited diagnostics. Frozen RADIO/SigLIP2/DINOv3/SAM3 heads supervise feature, relation, geometry, and mask-logit consistency without training task-specific evaluators; a feature-only prompt-conditioned SAM3 boundary head is additionally trained from official SAM3 training-view pseudo masks and evaluated without RGB SAM3 calls.

using rendered guidance signals such as visibility, alpha, and geometric depth.

3.3. Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^\star \rangle}{\|\hat{F}_v\|_2 \|F_v^\star\|_2} + \lambda_{rec} \|\hat{F}_v - F_v^\star\|_1. \quad (3)$$

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc} \mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \quad (4)$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

3.4. Frozen Adaptor Consistency

C-RADIOv4-H exposes frozen adaptor heads for `siglip2-g`, `dino_v3`, and `sam3`. The main open-vocabulary evaluator uses the text-aligned `siglip2-g` path, while training can additionally supervise the rendered feature through the other adaptors. For an adaptor A_a, we add a pointwise consistency term:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^\star)). \quad (5)$$

For `dino_v3`, we also match the token self-similarity structure rather than only the local feature value. Given normalized adaptor tokens D and $D^\star$, we compute:

$$\mathcal{L}_{rel} = \|DD^\top - D^\star D^{\star\top}\|_2^2. \quad (6)$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. We also evaluate a cross-view variant inspired by ProFuse's use of dense cross-view correspondence and context fusion [2]. For two training views p and q, we match teacher and rendered DINO token affinity:

$$\mathcal{L}_{cv} = \|D_p D_q^\top - D_p^\star D_q^{\star\top}\|_2^2. \quad (7)$$

Unlike ProFuse, this is an optimization-time regularizer on rendered feature maps, not a full pre-registration or context-proposal pipeline. To prevent cross-view smoothing from erasing text-grounding peaks, we add a label-free SigLIP text-heatmap distillation term over a fixed LERF text bank. The per-pixel query variant preserves the scene-category distribution:

$$\mathcal{L}_{txt}^{query} = \mathrm{KL}(\mathrm{softmax}(S^{\star}T^{\top}/\tau) \,\|\, \mathrm{softmax}(\hat{S}T^{\top}/\tau)). \tag{8}$$

We also evaluate a spatial variant that normalizes each query over image locations, directly targeting the LERF argmax localization failure mode:

$$\mathcal{L}_{txt}^{spatial} = \mathrm{KL}(\mathrm{softmax}_{xy}(s_q^{\star}/\tau) \,\|\, \mathrm{softmax}_{xy}(\hat{s}_q/\tau)). \tag{9}$$

For `sam3`, the current implementation uses a mask-free soft-region loss. Teacher SAM3 tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by teacher similarity, and the rendered feature is trained to match the teacher region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\mathrm{sam3}}(\hat{F}_v)) - \mu_k(A_{\mathrm{sam3}}(F_v^{\star}))\|_2^2. \tag{10}$$

This approximates SAM-style region supervision when external SAM mask caches are not available, while keeping the inference-time representation unchanged. We additionally implement a mask-logit distillation fallback for the `sam3` adaptor. Teacher SAM3-adaptor tokens define deterministic mask anchors, and we match the rendered feature's per-token soft assignment logits to the teacher distribution:

$$\mathcal{L}_{mask} = \mathrm{KL}\Big(\mathrm{softmax}(A_{\mathrm{sam3}}(F_v^{\star})P^{\top}/\tau) \,\|\, \mathrm{softmax}(A_{\mathrm{sam3}}(\hat{F}_v)P^{\top}/\tau)\Big), \tag{11}$$

where P are teacher anchor tokens. This remains an adaptor-space training loss rather than an official frozen SAM3 mask-decoder loss. The official SAM3 weights are used only in the separate promptable readout/cache path, where the decoder is frozen and does not train the evaluator.

3.5. Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

3.6. Open-Vocabulary Grounding

At evaluation time, CTF-GS renders a feature map for each novel view. For a text query q, we compute a text embedding e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ. The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center readout decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_{\theta}(c_i), \qquad s_i(q) = \cos(A_{\mathrm{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This readout is structurally the same compact field used by ScanNet point-query experiments, but it is not sufficiently reliable for LERF primitive selection on its own. We therefore introduce *View-to-Primitive Registration* (VPR) for the LERF direct-selection protocol. VPR renders teacher-space features from posed views, applies VFA and the frozen SigLIP2 head, registers those text-aligned features back to visible Gaussian centers with depth/alpha checks, and then runs the same primitive-level text query on the registered Gaussian embeddings. The promoted readout uses a fixed global score threshold with floor/cap bounds; voxel context and component grouping are kept as diagnostics. The LERF masks are used only for final query-select-render evaluation, not for registration, calibration, or scoring. To make the direct compact field itself more usable, we additionally train a label-free VPR-to-field consistency variant. Let $\bar{e}_i^{\mathrm{vpr}}$ be the cached normalized summary feature registered to Gaussian i by VPR, and let n_i be the number of VPR views that registered valid evidence to Gaussian i. The promoted transfer uses a GT-free confidence weight $w_i = \log(1+n_i)/\mathrm{mean}_j \log(1+n_j)$ over valid registered primitives. We fine-tune the compact field and a lightweight summary adaptor with

$$\mathcal{L}_{v2f} = 1 - \cos(A_{\mathrm{SigLIP2}}(D_{\theta}(c_i)), \mathrm{sg}[\bar{e}_i^{\mathrm{vpr}}]) + \lambda\,\mathcal{L}_{text},$$

where $\mathcal{L}_{text}$ matches the registered VPR scene-query distribution and uses only the LERF text categories, not object masks; all loss terms are averaged with w_i. This stage is a direct-field transfer objective: the streamed registered VPR readout remains the strongest paper-facing direct-3D result, while the VPR-to-field variant tests whether registered multiview evidence can be compressed back into the queryable Gaussian-center field. For the Dr. Splat-inspired audit, we also implement rasterizer-level Gaussian–pixel hit registration variants: all footprint contributions, per-pixel dominant hits, and per-Gaussian top-contribution pixels. These variants capture real rasterization intersections instead of only projected centers, but they are reported as diagnostics because they reduce the current LERF direct-3D metrics under the fixed protocol.

4. Experiments

4.1. Submission-Freeze Protocol

The conservative RADIO-GS numbers come from the 2026-05-02 submission-freeze report: `output/radio_gs/reports/submission_freeze_report.m` The RADIO adaptor ablations were completed on 2026-05-03 under the same LERF sweep protocol. External baseline rows are now official-source values from the cited papers or supplements and are captioned as cross-paper context rather than reproduced same-evaluator results; the

Table 1. Frozen CTF-GS LERF-OVS rendered-feature grounding result. mIoU uses a GT-free peak-relative mask threshold of 0.60 selected by the global boundary-calibration sweep in Table 3; LocAcc is unchanged from the heatmap argmax.

Scene	LocAcc	mIoU	Temp.
Figurines	0.8214	0.4244	50
Ramen	0.9014	0.6201	40
Teatime	0.8983	0.5760	25
Waldo Kitchen	0.8636	0.4769	25
Macro	0.8712	0.5243	–

Table 2. LERF-OVS open-vocabulary grounding comparison. External rows are official-source LocAcc values converted to decimals; CTF-GS is the local rendered-feature result. Best per column is in **bold**.

Method	Figurines	Ramen	Teatime	Waldo Kitchen	Macro
LERF	0.795	0.625	**0.938**	0.815	0.793
LangSplat	0.804	0.732	0.881	**0.955**	0.843
LEGaussians	0.767	0.737	0.683	0.523	0.678
RADIO-GS	**0.821**	**0.901**	0.898	0.864	**0.871**

exact source rows and protocol caveats are recorded in `baseline_source_verification.md`.

4.2. LERF-OVS Main Result

Table 1 is the current main internal result. Ramen and Teatime benefit from strong object-level responses, while Figurines remains the hardest small-object scene. The default threshold-0.50 readout gives 0.4941 macro mIoU; the global threshold-0.60 calibration tightens mask boundaries and raises the paper-facing macro mIoU to 0.5243 without changing localization accuracy. The adaptor-enhanced candidate in Table 20 is reported as an ablation/selector result rather than replacing the calibrated main row.

Table 2 provides the paper-facing external comparison. The LERF, LangSplat, and LEGaussians rows are traced to official paper or supplement tables and converted to decimal LocAcc when originally reported as percentages. Because these rows are not rerun under the RADIO-GS evaluator, we use them as context for the main claim rather than as a strict reproduced-SOTA leaderboard.

Table 3 isolates the rendered-mask boundary readout from feature quality. Raising the GT-free peak-relative mask threshold from 0.50 to 0.60 tightens the predicted region and improves macro mIoU on the paper-facing checkpoints from 0.494 to 0.524 without changing LocAcc. A threshold of 0.65 is marginally higher in macro mIoU but has lower weighted mIoU and visibly over-shrinks Figurines, so we use 0.60 as the safer global calibration for the main table and rendered qualitative masks. RGB boundary snap is more useful for direct 3D selection, where the rendered binary mask is only the final evaluation projection of already selected 3D primitives. An adaptive mean+std threshold diagnostic keeps LocAcc unchanged but lowers macro mIoU to 0.4939, so it is not promoted. We also implement a feature-only prompt-conditioned SAM3 boundary readout trained from official SAM3 pseudo masks on unlabelled training views. To avoid turning it into an RGB SAM3 evaluation-time oracle, the readout receives only rendered CTF-GS features, the SigLIP2 prompt, and the heatmap-threshold coarse mask. A query-support gate is required: the refined mask must substantially overlap the original heatmap support, obey bounded area change, and is clipped to a small dilation band around the query support. Under the current active-checkpoint run, the stable global gate raises weighted sample mIoU from 0.5493 to 0.5666 without changing LocAcc and removes the previous Ramen regression. Because the scene-category macro remains lower at 0.5141 mIoU, compared with 0.5511 scene-macro and 0.5666 sample-weighted mIoU, we keep it as a boundary-readout ablation rather than replacing Table 1.

Table 3. GT-free rendered-mask boundary calibration on the paper-facing LERF-OVS checkpoints. All rows use the same checkpoints, text embeddings, temperatures, scoring, heatmap upsampling, and rendered heatmaps; only the peak-relative binary mask threshold changes. LocAcc is unchanged because it is measured from the heatmap argmax before binarization.

Readout	Fig.	Ramen	Tea.	Waldo	Macro
Threshold 0.50	**0.431**	0.586	0.549	0.411	0.494
Threshold 0.55	0.430	0.611	0.568	0.451	0.515
Threshold 0.60	0.424	**0.620**	0.576	0.477	0.524
Threshold 0.65	0.415	0.619	**0.578**	**0.490**	**0.525**
Threshold 0.70	0.395	0.596	0.556	0.477	0.506

4.3. LERF Direct 3D Object Selection

Table 4 evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table 1; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

The original Gaussian-center readout reached 0.080 macro mIoU and 0.093 macro Acc@0.25 under an earlier top-10% selector, but only 0.012/0.009 under the same fixed top-2% selector used by the VPR ablation rows. The registration readout closes most of this gap without using LERF masks for scoring. Under the same VPR readout,

Table 4. LERF-OVS direct 3D object selection under an OpenGaussian-style query-select-render protocol. External values are the OpenGaussian official paper numbers. The VPR row uses registered primitive features, a fixed global threshold, and GT-free RGB snap. The compact-field row uses deployed Gaussian-center primitive scores with an opacity-gated point adapter and does not read a VPR cache at inference. The SAM3-box fixed row scores compact direct-field primitives, renders the selected 3D primitives, and uses an official SAM3 box-prompt readout selected by overlap with the rendered mask, not by ground truth. The scene-locked SAM3-box row is diagnostic.

Method	Protocol	Fig.	Ramen	Tea.	Waldo	Macro
OpenGaussian	mIoU	0.393	0.310	0.604	0.227	0.384
CTF-GS	thr0p25 mIoU	0.531	0.580	0.566	0.243	0.480
CTF-GS compact	thr0p35 mIoU	0.515	0.573	0.548	0.299	0.484
CTF-GS	SAM3 box fixed mIoU	0.614	0.641	0.613	0.414	0.570
CTF-GS	SAM3 box scene-locked mIoU	0.642	0.649	0.653	0.444	0.597
CTF-GS	diag. best mIoU	0.533	0.582	0.566	0.246	0.482
OpenGaussian	Acc@0.25	0.554	0.422	0.763	0.318	0.514
CTF-GS	thr0p25 Acc@0.25	0.786	0.746	0.763	0.409	0.676
CTF-GS compact	thr0p35 Acc@0.25	0.661	0.789	0.712	0.409	0.643
CTF-GS	SAM3 box fixed Acc@0.25	0.696	0.746	0.746	0.546	0.684
CTF-GS	SAM3 box scene-locked Acc@0.25	0.732	0.746	0.780	0.546	0.701

the mean+std score-distribution selector with a 0.5% floor, 1.8% cap, and GT-free RGB snap reaches 0.446/0.661. Replacing it with a single global softmax-score threshold of 0.25 reduces primitive clutter and improves the promoted row to 0.480 macro mIoU and 0.676 macro Acc@0.25. The compact direct-field row with official SAM3 box-prompt readout refinement is reported with a fixed global selector: the pad16 'thr0p25' readout reaches 0.570 macro mIoU and 0.684 macro Acc@0.25, while the scene-locked diagnostic upper bound reaches 0.597/0.701. SAM3 is used only after the 3D primitives have been scored and rendered; its candidate mask is selected by overlap with the rendered prediction rather than ground truth. This is above the OpenGaussian official macro reference while using a different SigLIP2 text head and no locally rerun baseline reproduction. In the OpenGaFF-aligned published context, the same fixed SAM3-box row exceeds the reported direct-3D mIoU context value but trails the reported Acc@0.25. Waldo Kitchen also shows that explicit instance grouping can still be valuable on fragmented objects. We therefore use the result as evidence for direct-field usability with a frozen promptable boundary readout rather than as a universal primitive-level SOTA claim.

Table 5 gives the OpenGaFF-aligned published context requested by the protocol audit. These rows are deliberately separated from the local result table: they mix papers, text heads, and implementation details, so they motivate positioning against recent primitive-/instance-aware methods rather than serving as a strict same-evaluator leaderboard.

Table 6 makes the direct-3D protocol auditable. The registration step uses 128 posed rendered views and visibility checks, but no LERF object masks or query labels. The VPR paper-facing selector uses a fixed global softmax threshold with fixed floor/cap bounds. The SAM3-box paper-facing boundary readout also reports a fixed global threshold. Fixed-ratio sweeps, best-by-scene choices, SAM3-box scene-locked threshold exports, and component-refinement variants are diagnostic until validation-selected.

Table 7 audits the implementation choices and disambiguates the two raw-center diagnostics: 0.080/0.093 is the earlier top-10% selector, whereas 0.012/0.009 is the same top-2% selector used by the VPR ablation rows. Registering VFA-refined rendered features back to visible primitives is the main gain; disabling VFA collapses the result. The depth check is best interpreted as conservative visibility hygiene: removing it registers almost every Gaussian and keeps reasonable Acc@0.25, but does not improve mIoU over the promoted 96-view VPR row. The train-view-only row is a conservative leakage audit: it remains well above the raw center readout without using valida-

Table 5. OpenGaFF-aligned published-context LERF-OVS direct 3D object-selection numbers. Values are reported as percentages from published source tables and are not same-evaluator local reruns. This table is only for positioning relative to recent primitive-/instance-level methods with publicly reported results.

Method	Source type	mIoU	Acc@0.25
OpenGaussian [16]	official paper	38.36	51.43
Dr. Splat [6]	published context	43.29	64.30
InstanceGaussian [7]	published context	45.30	58.44
OpenGaFF [8]	arXiv context	54.36	80.84
CTF-GS + VPR	local, SigLIP2, fixed `thr0p25` + RGB snap	48.01	67.60
CTF-GS + direct field + SAM3 box	local, fixed global threshold, frozen official SAM3 readout	57.05	68.35
CTF-GS + direct field + SAM3 box	local diagnostic, scene-locked thresholds	59.72	70.09

Table 6. VPR protocol card for LERF-OVS direct 3D object selection.

Item	Setting
Query target	3D Gaussian primitives; rendering is used only after selection for mask evaluation
Registration feature	Rendered RADIO-compatible features after VFA, projected by frozen SigLIP2 summary head
Registration views	Posed RGB/render views selected from all COLMAP poses; main row uses 128 evenly spaced views
Leakage audit	Train-view-only 24-view row is reported separately and remains above raw center readout
Visibility hygiene	Depth and alpha checks; no LERF masks or text labels are used during registration
Text scoring	Frozen SigLIP2 text/query embeddings with scene-wise softmax scoring
Selection	Fixed global softmax-score threshold `thr0p25` with 0.5% floor and 1.8% cap; mean+std and fixed-ratio sweeps are diagnostic only
Context aggregation	GT-free voxel-max score aggregation, resolution 80, blend 0.50
Persistent storage	No extra trained state; optional registered primitive/voxel caches are reported separately
Metrics	OpenGaussian-style mIoU and Acc@0.25 after rendering selected primitives

Table 7. VPR diagnostics for LERF-OVS direct 3D object selection. Except for the explicitly marked historical raw-center row, rows use the same fixed top-2% primitive selector. "Reg." is the mean fraction of Gaussians receiving registered rendered-view features; unregistered primitives use the direct Gaussian-center fallback unless disabled by the ablation.

Variant	Selector / views	Reg.	mIoU	Acc@0.25
Raw Gaussian-center	top10%, previous	–	0.080	0.093
Raw Gaussian-center	top2%, VPR protocol	–	0.012	0.009
VPR, cosine	top2%, all poses 24	0.385	0.312	0.547
VPR, softmax	top2%, all poses 24	0.385	0.342	0.555
VPR + voxel	top2%, train views 24	0.369	0.349	0.595
VPR + voxel	top2%, all poses 8	0.240	0.277	0.487
VPR + voxel	top2%, all poses 48	0.448	0.380	0.653
VPR + voxel	top2%, all poses 96	0.485	0.385	0.643
VPR + voxel, lower blend	top2%, all poses 96	0.485	0.385	0.631
VPR + voxel, w/o VFA	top10%, all poses 24	0.385	0.070	0.099
VPR + voxel, w/o depth check	top2%, all poses 24	0.972	0.347	0.615

Table 8. VPR-to-field consistency for LERF-OVS direct 3D object selection. All rows use the same softmax-threshold selector family, 0.5% floor, 1.8% cap, and GT-free RGB boundary snap. Raw and VPR-to-field rows report per-scene diagnostic-best thresholds from the sweep; registered VPR reports the fixed thr0p25 paper row. VPR-to-field distills cached registered VPR summary features into the compact direct field and evaluates the direct Gaussian-center readout with a 0.5 residual blend to the original field. The confidence-weighted row uses GT-free log view-count weights from VPR registration.

Readout	Metric	Fig.	Ramen	Tea.	Waldo	Macro
Raw Gaussian-center	mIoU	0.001	0.006	0.158	0.019	0.046
VPR-to-field consistency	mIoU	0.488	0.438	0.510	0.212	0.412
VPR-to-field + conf. weights	mIoU	0.404	0.598	0.520	0.225	0.436
Registered VPR readout	mIoU	0.531	0.580	0.566	0.243	0.480
Raw Gaussian-center	Acc@0.25	0.000	0.000	0.237	0.045	0.071
VPR-to-field consistency	Acc@0.25	0.714	0.578	0.695	0.364	0.588
VPR-to-field + conf. weights	Acc@0.25	0.643	0.746	0.678	0.409	0.619
Registered VPR readout	Acc@0.25	0.786	0.746	0.763	0.409	0.676

tion or annotated frames for registration. We also tested Dr. Splat-inspired contribution-weighted registration. The earlier center-sampled alpha and alpha-depth variants lowered the fixed-protocol macro result to 0.298 and 0.297 mIoU, respectively. The new rasterizer-level variants are also negative on Figurines under the same paper selector: all-footprint raster hits with uniform weights reach only 0.0002 mIoU, per-pixel dominant alpha-depth hits reach 0.0178 mIoU with the 128-view budget, and per-Gaussian top-footprint alpha hits reach 0.0004 mIoU in the official-view probe. Proposal/OPR component selection on the cached strong VPR scores also drops Figurines to 0.0430 mIoU. We therefore keep center-based uniform VPR with global score-threshold selection as the promoted row and report raster/proposal branches as implemented negative diagnostics rather than paper-facing improvements. We also audited GT-free confidence selectors based on score margin, ratio, and entropy. The best margin sweep reaches 0.446 macro mIoU and 0.628 macro Acc@0.25, below the fixed global-threshold row, so confidence selection is kept as a diagnostic rather than the main protocol.

Table 8 tests whether the VPR signal can be compressed back into the student field. Under the same threshold-sweep diagnostic selector family and boundary snap, raw direct Gaussian-center scoring is nearly unusable at 0.0458 macro mIoU and 0.0707 Acc@0.25. VPR-to-field consistency lifts this direct-field readout to 0.4119/0.5876, and GT-free confidence weighting further improves it to 0.4363/0.6191, closing much of the gap to the streamed registered VPR readout at 0.4799/0.6760. This supports the student foundation-feature-field claim more directly than a pure registration cache, while the remaining gap, especially on Waldo Kitchen, keeps registered VPR as the promoted direct-3D readout.

Table 9 audits the promoted direct-3D selector at query level. Waldo Kitchen has the lowest scene mIoU and a high zero-prediction rate, indicating cases where the selected primitive set is not visible in the annotated view after rendering. Ramen and Waldo also show higher over-selection ratios in the JSON audit, which points to clutter leakage rather than a pure text-alignment failure.

Table 9. Query-level audit for LERF direct 3D object selection. Confidence intervals are bootstrap intervals over query masks for the fixed thr0p25 selector.

Scene	Queries	mIoU	Acc@0.25	Zero pred.
Figurines	56	0.531	0.786	0.071
Ramen	71	0.580	0.746	0.028
Teatime	59	0.566	0.763	0.017
Waldo Kitchen	22	0.243	0.409	0.182

Table 10. Direct3D mechanism audit using GT-free VPR view coverage, primitive teacher-score confidence, and text-margin ambiguity. Scene rows report mean valid registered views; teacher-bucket rows report mean top-1% primitive text score; text-margin rows report score margin against the strongest competing category.

Group	Item	Support	Coverage/score	mIoU
Scene	figurines	56	9.5911	0.5309
Scene	ramen	71	20.8716	0.5805
Scene	teatime	59	13.0006	0.5662
Scene	waldo_kitchen	22	5.7542	0.2429
Teacher bucket	low	70	0.1800	0.4345
Teacher bucket	mid	69	0.4257	0.5132
Teacher bucket	high	69	0.6939	0.6358
Text margin	ambiguous	70	-0.0882	0.4329
Text margin	mixed	69	0.2451	0.5303
Text margin	distinct	69	0.6074	0.6202

Table 10 adds a GT-free mechanism view over the same Direct3D result. Scene-level mean valid VPR views correlate with strict direct-selection mIoU at Pearson $r = 0.7588$, while high teacher-score query instances reach 0.6358 mean IoU and 0.8551 Acc@0.25 versus 0.4345 and 0.6143 in the low-score bucket. Text-margin stratification shows the same pattern: distinct-margin queries reach 0.6202/0.8261, while ambiguous queries are 0.4329/0.6286. This supports treating Waldo Kitchen as a coverage/confidence/ambiguity bottleneck in addition to a boundary-fragmentation failure.

Table 11 measures the boundary side of the strict SAM3-box readout. The pad16 fixed-threshold row has macro Boundary-F 0.6681 and trimap IoU 0.3958, and the 208 per-query records show that IoU is strongly correlated with Boundary-F ($r = 0.9148$) and trimap IoU ($r = 0.8412$). Balanced predicted/GT mask sizes have much

Table 11. Measured boundary-error readout for the strict SAM3-box Direct3D row. Boundary error is 1−Boundary-F and trimap error is 1−trimap IoU.

Scene	mIoU	Boundary-F	Boundary err.	Trimap IoU	Trimap err.
Figurines	0.6136	0.7116	0.2884	0.3986	0.6014
Ramen	0.6409	0.7713	0.2287	0.4400	0.5600
Teatime	0.6130	0.7255	0.2745	0.4626	0.5374
Waldo Kitchen	0.4142	0.4638	0.5362	0.2820	0.7180

Table 12. Alpha/depth discontinuity alignment coverage for the strict SAM3-box Direct3D row. Geometry records count per-query alpha/depth edge metrics and overlays.

Scene	Queries	Geometry records	mIoU	Boundary-F
Figurines	56	56	0.6136	0.7116
Ramen	71	71	0.6409	0.7713
Teatime	59	59	0.6130	0.7255
Waldo Kitchen	22	22	0.4142	0.4638

Table 13. Same-evaluator comparison between original RADIO features extracted from RGB frames and CTF-GS rendered features. Both use the same frozen SigLIP2 head, scene-category scoring, and GT-free threshold-0.60 mask readout.

Scene	RADIO RGB		CTF-GS	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7321	0.3816	0.8214	0.4244
Ramen	0.8873	0.5247	0.9014	0.6201
Teatime	0.8475	0.5192	0.8983	0.5760
Waldo Kitchen	0.7273	0.4280	0.8636	0.4769
Macro	0.7985	0.4634	0.8712	0.5243

lower mean boundary error (0.0637) than under-selected or over-selected masks (0.7042 and 0.6919), so the failure is better described as mask-size and boundary alignment breakdown than as a pure text-score issue. The evaluator now supports saving per-query alpha/depth discontinuity overlays for this analysis. The strict pad16 geometry-map rerun populates 208/208 query records; because the alpha/depth correlations with boundary error are weak, we use these overlays as mechanism diagnostics rather than causal occlusion evidence.

4.4. Rendered Features vs. Original RADIO Features

Table 13 supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of the teacher. It can sharpen the task-relevant scene response after multiview reconstruction: under the same calibrated mask readout it improves both localization and macro mIoU over frame-wise RADIO RGB features, even though the original RADIO RGB feature remains the direct 2D teacher used during training.

Table 14. Nearest-view cached-teacher baseline on LERF-OVS. Each annotated target frame uses the closest cached RADIO feature map by camera-center distance, excluding the target frame itself; the feature map is not warped into the target camera.

Scene	LocAcc	mIoU	N	Dist.
Figurines	0.1786	0.0755	56	0.2831
Ramen	0.2817	0.1740	71	0.5443
Teatime	0.3559	0.2114	59	0.4266
Waldo Kitchen	0.2727	0.1570	22	0.5787
Macro	0.2722	0.1545	–	0.4582

Table 15. Per-Gaussian 1280-D explicit RADIO-memory baseline on LERF-OVS. Cached teacher features are registered to Gaussian centers, stored as fp16 vectors, rendered to annotated views, and evaluated with the same frozen SigLIP2 scorer.

Scene	LocAcc	mIoU	Reg. frac.	Storage MiB
Figurines	0.6429	0.2729	0.1123	412.1
Ramen	0.6056	0.4333	0.2835	934.3
Teatime	0.5085	0.3541	0.2647	1123.4
Waldo Kitchen	0.5000	0.2125	0.1475	1688.8
Macro	0.5642	0.3182	0.2020	1039.7

Table 14 adds a stricter cache-only control: for each annotated target frame, it substitutes the nearest cached RADIO feature frame by camera-center distance, excludes the target frame itself, and performs no geometric warping. The resulting 0.2722 macro LocAcc and 0.1545 macro mIoU are far below the rendered CTF-GS feature row in Table 13. This separates the 3D reconstructed feature-field contribution from a simple 2D nearest-view feature cache.

Table 15 adds the raw-feature 3D memory control requested by the controlled-evidence audit. Cached RADIO teacher features are registered to visible Gaussian centers, stored as fp16 1280-D vectors, rendered to the annotated LERF views, and scored with the same frozen SigLIP2 readout. The baseline reaches 0.5642 macro LocAcc and 0.3182 macro mIoU with 0.2020 mean registered-Gaussian fraction and 1039.7 MiB mean feature storage. This is stronger than the cache-only row but remains below compact CTF-GS (0.8712 / 0.5243), supporting the need for learned compact reconstruction rather than direct sparse teacher-feature attachment.

Table 16 links global feature reconstruction quality to the frozen text-grounding readout. Across the four LERF scenes, $1 - \cos_{decoded}$ correlates with rendered mIoU error at $r = 0.9568$ and with LocAcc error at $r = 0.8713$. This

Table 16. Feature reconstruction error versus text relevance error. Feature error is 1− best validation decoded cosine from the scene training log; text relevance errors are derived from the frozen rendered-grounding readout.

Scene	Feature err.	mIoU err.	LocAcc err.	Best cosine
Figurines	0.2382	0.5756	0.1786	0.7618
Ramen	0.1624	0.3799	0.0986	0.8376
Teatime	0.2005	0.4240	0.1017	0.7995
Waldo Kitchen	0.2209	0.5231	0.1364	0.7791

Table 17. Controlled LERF-OVS component ablation on seed 7. The table reports rendered-feature LocAcc per scene plus macro LocAcc and mIoU.

Variant	Fig.	Ramen	Tea.	Waldo	LocAcc	mIoU
Full CTF-GS	0.768	0.901	0.898	0.864	0.858	0.485
w/o FGC warm-start	0.696	0.845	0.847	0.818	0.802	0.424
w/o VFA	0.768	0.859	0.915	0.818	0.840	0.480
w/o HGCF	0.768	0.873	0.898	0.818	0.839	0.507
w/o CTR	0.268	0.648	0.661	0.545	0.531	0.260

Table 18. Storage footprint accounting for compact teacher-feature fields and optional VPR caches. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian. Compact ckpt counts the stored feature-field model, CTR/HCD codec, and VFA/refiner tensors. VPR caches are inference-time artifacts, not persistent trained state.

Scene	#G	Direct MiB	Compact MiB	Saving	Opt. VPR cache MiB	Saving w/ cache
Figurines	168,791	412.1	237.0	1.74×	501.3	0.56×
Ramen	382,687	934.3	311.2	3.00×	1131.4	0.65×
Teatime	460,157	1123.4	338.1	3.32×	1360.4	0.66×
Waldo Kitchen	691,728	1688.8	418.5	4.04×	2050.3	0.68×

Table 19. Compression ratio versus downstream mIoU. Saving ratio compares direct per-Gaussian 1280-D fp16 teacher-feature storage with the compact checkpoint footprint; downstream metrics use the frozen rendered-grounding and Direct3D readouts.

Scene	Saving	Rendered mIoU	Direct3D mIoU	Direct3
Figurines	1.74×	0.4244	0.5309	
Ramen	3.00×	0.6201	0.5805	
Teatime	3.32×	0.5760	0.5662	
Waldo Kitchen	4.04×	0.4769	0.2429	

supports the feature-reconstruction thesis, while the small scene count keeps it a mechanism audit rather than a per-query causal proof.

4.5. Core Component Ablation

Table 17 isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table 1, which reports the conservative current best per scene, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: HGCF tests compact feature storage, CTR tests the nonlinear compact-to-teacher bottleneck, VFA tests peak and boundary correction, and FGC warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU (0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy on Ramen/Waldo. This explains why some enhancement branches can improve mIoU without preserving LocAcc.

4.6. Storage Footprint

Table 18 closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian and excludes ordinary RGB/geometry attributes. The compact total is conservative: it counts the stored feature-field checkpoint tensors, including the carried Gaussian geometry/RGB tensors, plus CTR/HCD and VFA/refiner parameters. Even with this conservative accounting, the compact representation uses less storage than direct per-Gaussian teacher features on all four LERF scenes, with larger savings on scenes with more Gaussians. VPR itself is an inference-time readout and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature storage. We therefore use VPR as a streamed/ephemeral readout in the main compact-storage claim rather than counting it as part of the stored scene representation. Table 19 further separates compactness from robustness: checkpoint saving ratio has only weak positive correlation with rendered mIoU ($r = 0.3606$) and negative correlation with Direct3D mIoU ($r = -0.6158$). This prevents overclaiming compression as the source of better querying; Waldo Kitchen has the largest storage saving but remains the weakest direct-3D scene.

4.7. Adaptor Consistency Ablation

Table 20 shows that adaptor supervision is useful but scene-dependent under the default threshold-0.50 readout. DINO

Table 20. Full-sweep LERF-OVS adaptor ablations under the original threshold-0.50 readout. We only promote adaptor variants that preserve localization accuracy while improving mIoU; these diagnostics are kept separate from the calibrated Table 1 result.

Scene	Variant	LocAcc	mIoU
Figurines	baseline	0.8214	0.4308
Figurines	DINO rel. + SAM3 region	0.8036	0.4399
Figurines	DINO cv + spatial text	0.8214	0.4343
Ramen	baseline	0.9014	0.5862
Ramen	DINO rel. + SAM3 region	0.9014	0.5873
Teatime	baseline	0.8983	0.5486
Teatime	DINO rel. + SAM3 region	0.8983	0.5592
Waldo Kitchen	baseline	0.8636	0.4106
Waldo Kitchen	DINO + SAM3 pointwise	0.8636	0.4103
Macro	promoted selector	0.8712	0.4979

relation plus SAM3 region supervision improves Ramen without hurting localization. Teatime also benefits from the relation/region variant, reaching 0.5592 mIoU at unchanged localization accuracy. For Figurines, DINO cross-view with spatial text-heatmap distillation recovers the baseline localization score while improving mIoU from 0.4308 to 0.4343. Waldo Kitchen remains on the baseline checkpoint. Using the promoted Figurines, Ramen, and Teatime adaptor/cross-view checkpoints while keeping baseline Waldo Kitchen gives a candidate adaptor-enhanced macro score of 0.8712 LocAcc and 0.4979 mIoU. Since the global boundary calibration in Table 3 supersedes this readout for the main LERF mask metric, adaptor consistency remains a method diagnostic rather than the promoted quantitative row.

The main failure mode is a localization/region tradeoff. The DINO relation and SAM3 region losses often smooth or broaden object-consistent heatmaps, which can raise thresholded overlap while moving the single argmax outside a small polygon. Because LERF-OVS LocAcc is an argmax-inside-mask metric, one feature-cell shift can hurt LocAcc even when the region response is visually cleaner. For submission, we therefore only promote adaptor checkpoints that preserve LocAcc; future training should add an explicit peak-preservation term or mask-aware small-object weighting before replacing the conservative main table.

Table 21 answers the cross-view question directly. DINO cross-view affinity and text-heatmap preservation can increase thresholded overlap, especially on Waldo at high temperature. The spatial heatmap variant recovers the Figurines localization score and becomes the only promoted cross-view LERF branch, but Waldo still loses the single-peak localization score. This suggests that cross-view affinity encourages coherent object regions while peak placement remains scene dependent; a full ProFuse-style pre-registration/context proposal module remains a future

Table 21. ProFuse-inspired DINOv3 cross-view diagnostics on LERF-OVS under the original threshold-0.50 readout. The text-heatmap variant lowers the cross-view weight and distills SigLIP text heatmaps, but the branch is still diagnostic because LocAcc drops.

Scene / Variant	Temp.	LocAcc	mIoU
Figurines baseline	50	0.8214	0.4308
Figurines DINO cross-view	50	0.8036	0.4339
Figurines cross-view + text heatmap	50	0.8036	0.4381
Figurines cross-view + spatial heatmap	50	0.8214	0.4343
Waldo baseline	25	0.8636	0.4106
Waldo DINO cross-view	25	0.5909	0.3662
Waldo cross-view + text heatmap	40	0.6818	0.4563

Table 22. Frozen adaptor-space downstream probes on LERF-OVS. "Teacher" uses original RADIO RGB features; "Rendered" uses CTF-GS rendered features. Prototype segmentation uses same-category support masks from other frames. Matching uses the first annotated frame as source.

Adaptor	Probe	Teacher		Rendered	
		LocAcc	mIoU	LocAcc	mIoU
DINOv3	Prototype seg.	0.6543	0.0945	0.6277	0.0937
DINOv3	Source-target match	0.5957	0.1032	0.5035	0.1019
SAM3	Prototype seg.	0.8404	0.0757	0.6649	0.0564
SAM3	Source-target match	0.7872	0.0953	0.7092	0.0687

method extension rather than a paper-main result.

4.8. DINO/SAM Adaptor Downstream Probes

Table 22 is diagnostic rather than a claimed improvement: rendered features are competitive in DINOv3 mIoU but do not yet beat original RADIO features on aggregate, and unconstrained SAM3 prototype scoring exposes a larger teacher-rendered gap. Per-scene positives still exist, especially on Waldo Kitchen source-target matching, where rendered DINOv3 improves from 0.2500 to 0.5000 LocAcc and from 0.2708 to 0.2948 mIoU, and rendered SAM3 improves mIoU from 0.2965 to 0.3147 at unchanged LocAcc. These probes therefore support the feature-compatibility story while identifying adaptor-space preservation as the next optimization target. We therefore also report prompt-constrained task probes that better match the intended SAM3-adaptor and DINOv3-adaptor downstream use cases.

Table 23 separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over the frame-wise teacher for point prompts, box prompts, and mask propagation, while preserving perfect point-prompt localization. Second, source-background contrast makes DINOv3 mask propagation substantially stronger; adding robust foreground pooling, area-bounded propa-

Table 23. Promptable SAM3-adaptor and dense DINOv3-adaptor tasks on LERF-OVS. These probes use the same annotations as the grounding benchmark but replace text queries with adaptor-space prompts or source-target correspondences. SAM3 point/box prompts are prompt-constrained, mask propagation uses source-view support only, and no external SAM3 mask decoder is called. DINO mask propagation uses source-background contrast 1.10 plus GT-free robust readout: top-k foreground pooling, 2.0$\times$ propagated-area scaling, and peak-component cleanup.

Task	Teacher		Rendered	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4169
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6667	0.3756
DINOv3 dense matching	0.5723	0.8547	0.5393	0.9048
DINOv3 mask propagation + bg-suppressed readout	0.7660	0.5119	0.7943	0.4805

gation, and connected-component cleanup raises rendered DINO mask-propagation from the previous robust row of 0.7730/0.4456 to 0.7943/0.4805 LocAcc/mIoU. This essentially matches the previous robust teacher mIoU reference of 0.4806 and exceeds the same-readout teacher LocAcc, but under the same background-suppressed readout the teacher remains higher in mIoU (0.5119). We therefore use this result as evidence that the DINO propagation gap is substantially narrowed, not as a same-protocol mIoU superiority claim. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants evaluated next. We also evaluate a mutual-correspondence plus homography-RANSAC diagnostic for qualitative DINO matching; it removes many outlier matches and raises rendered dense-match similarity to 0.9277, but it does not improve the same robust DINO mask-propagation mIoU beyond the teacher, so it remains a visualization and diagnostic filter rather than a main quantitative claim.

4.9. ScanNet Direct Point-Query Transfer

Table 24 evaluates cross-domain feature usability on the VALA/OpenGaFF ScanNet-8 scene subset: `scene0000_00`, `scene0062_00`, `scene0070_00`, `scene0097_00`, `scene0140_00`, `scene0347_00`, `scene0400_00`, and `scene0590_00`. This follows the OpenGaFF/VALA direct point-query setting with every-20-frame ScanNet training splits and mIoU/mAcc evaluation on GT semantic point clouds. The conservative block uses `gaussian_index` queries, `label_point` positions, and `label_index` opacity filtering. This is not presented as a full ScanNet semantic segmentation leaderboard result; it is a fair direct probe of the learned feature field. We also evaluate a label-free contextual point readout that queries each ScanNet vertex from nearby Gaussians instead of the row-aligned Gaussian index. On

Table 24. VALA/OpenGaFF ScanNet-8 teacher-balanced direct point-query results. Contextual rows use kNN point readout with $k = 8$, 32 candidate Gaussians, and label-free scene-mean logit calibration with $\alpha = 0.5$.

Readout	Split	mIoU	mAcc
Gaussian-index	19 classes	0.3583	0.6006
Gaussian-index	15 classes	0.3618	0.6152
Gaussian-index	10 classes	0.4367	0.6998
DINO-CV Gaussian-index	19 classes	0.3704	0.6159
DINO-CV Gaussian-index	15 classes	0.3718	0.6268
DINO-CV Gaussian-index	10 classes	0.4390	0.7020
Contextual kNN	19 classes	0.3677	0.5997
Contextual kNN	15 classes	0.3748	0.6181
Contextual kNN	10 classes	0.4562	0.7008
DINO-CV contextual kNN	19 classes	**0.3704**	0.6017
DINO-CV contextual kNN	15 classes	**0.3771**	0.6198
DINO-CV contextual kNN	10 classes	**0.4585**	**0.7032**

the original v67 field this kNN readout improves the three split mIoUs to 0.3677, 0.3748, and 0.4562 while keeping mAcc comparable. Combining the same readout with the DINO-CV compact field further improves the balanced row to 0.3704, 0.3771, and 0.4585 mIoU, with 0.6017, 0.6198, and 0.7032 mAcc. Increasing the calibration scale to 0.75 raises the 10-class mIoU to 0.4612 but lowers the 19-/15-class and mAcc averages, so we use $\alpha = 0.5$ as the balanced setting. ScanNet alias prompts remain mixed and are kept as appendix evidence. Table 25 gives the corresponding published-context comparison against the external rows reported by OpenGaFF v2. These external numbers are cited from that paper rather than reproduced locally. Under this protocol, the DINO-CV contextual CTF-GS row slightly exceeds the OpenGaFF published 19-class mIoU and is stronger than prior pre-OpenGaFF rows on mAcc, but remains below OpenGaFF on the 15-/10-class mIoU and 10-class mAcc. We therefore claim split-aligned transfer evidence rather than ScanNet leaderboard dominance.

Table 26 upgrades the earlier two-scene diagnostic to the VALA/OpenGaFF ScanNet-8 subset. DINO cross-view affinity improves the macro 19-class and 15-class probes more clearly than the 10-class probe, with positive macro mIoU and mAcc on all splits. The effect remains scene-dependent: the DINO branch improves 4/8, 5/8, and 5/8 scenes on the 19-, 15-, and 10-class mIoU splits, respectively. With the contextual direct readout in Table 24, it becomes the strongest balanced ScanNet support row. We also evaluated a stricter joint-readout training variant that aligns direct point features with the current rendered compact map and applies visibility-weighted text contrast. Its full ScanNet-8 result improves split10 mAcc but weakens split19/15 mIoU, so we keep it as an ablation rather than

Table 25. Published-context ScanNet-v2 open-vocabulary point-cloud understanding results under the OpenGaFF/VALA protocol. External rows are copied from the OpenGaFF v2 paper table and reported in percent; CTF-GS rows are our local VALA/OpenGaFF-8 evaluations on the same eight scene ids.

Method	19 mIoU	19 mAcc	15 mIoU	15 mAcc	10 mIoU	10 mAcc
LangSplat	2.45	8.59	3.45	13.21	6.48	21.89
LangSplatV2	14.75	25.47	17.09	35.68	22.83	41.52
OpenGaussian	27.73	42.01	29.67	46.15	39.93	57.34
Dr. Splat	29.31	47.68	33.25	54.33	44.19	65.19
OccamLGS	31.93	48.93	34.25	53.71	45.16	64.39
VALA	32.11	50.05	35.10	54.77	46.21	65.61
OpenGaFF	36.55	50.57	42.78	72.85	57.85	77.93
CTF-GS Gaussian-index	35.83	60.06	36.18	61.52	43.67	69.98
CTF-GS contextual kNN	36.77	59.97	37.48	61.81	45.62	70.08

Table 26. VALA/OpenGaFF ScanNet-8 DINOv3 cross-view ablation. The branch is warm-started from the v67 teacher-balanced checkpoint and uses conservative DINO cross-view affinity weight $\lambda = 0.001$ with batch size 2.

Split	Base mIoU	DINO mIoU	Δ	Base mAcc	DINO mAcc	Δ
19 classes	0.3583	0.3704	+0.0121	0.6006	0.6159	+0.0153
15 classes	0.3618	0.3718	+0.0100	0.6152	0.6268	+0.0116
10 classes	0.4367	0.4390	+0.0023	0.6998	0.7020	+0.0022

Table 27. Submission efficiency and cost evidence. Runtime rows use explicit freeze-profile telemetry; storage rows compare direct per-Gaussian 1280-D fp16 teacher features against the stored compact feature-field footprint.

Evidence	Scope	Cost	Peak VRAM
LERF eval	4 scenes	124.770 s / 31.193 s-scene	2076 MiB
ScanNet eval	10-scene superset	150.903 s / 15.090 s-scene	1666 MiB
Storage	4 scenes	1.74×–4.04× saving	

Table 28. Representative fragile LERF-OVS categories from the frozen rendered-feature evaluation. Low LocAcc with nonzero mIoU highlights the peak-vs-region trade-off.

Scene	Category	LocAcc	mIoU
Figurines	bag	0.00	0.00
Figurines	miffy	0.00	0.01
Teatime	dall-e brand	0.00	0.01
Figurines	jake	0.00	0.02
Waldo Kitchen	ketchup	0.00	0.03
Waldo Kitchen	spatula	0.00	0.03
Ramen	corn	0.00	0.10
Figurines	red toy chair	0.00	0.13

the main paper row.

4.10. Efficiency and Cost

Table 27 separates profiled evaluation cost from storage footprint accounting. The four LERF overlay evaluations take 124.770 seconds in total and peak at 2076 MiB, while the previously profiled 10-scene ScanNet v67 direct-query superset takes 150.903 seconds and peaks at 1666 MiB. Training throughput is kept as supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types.

4.11. Qualitative Results

The current qualitative shortlist is regenerated from the calibrated rendered mask readout used in Table 1. The recommended main figure uses one frozen rendered grid from each LERF scene: Figurines frame 00152, Ramen frame 00024, Teatime frame 00140, and Waldo Kitchen frame 00089.

4.12. Failure Analysis

Table 28 turns the qualitative small-object observation into a paper-facing diagnostic. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a plausible object region while the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

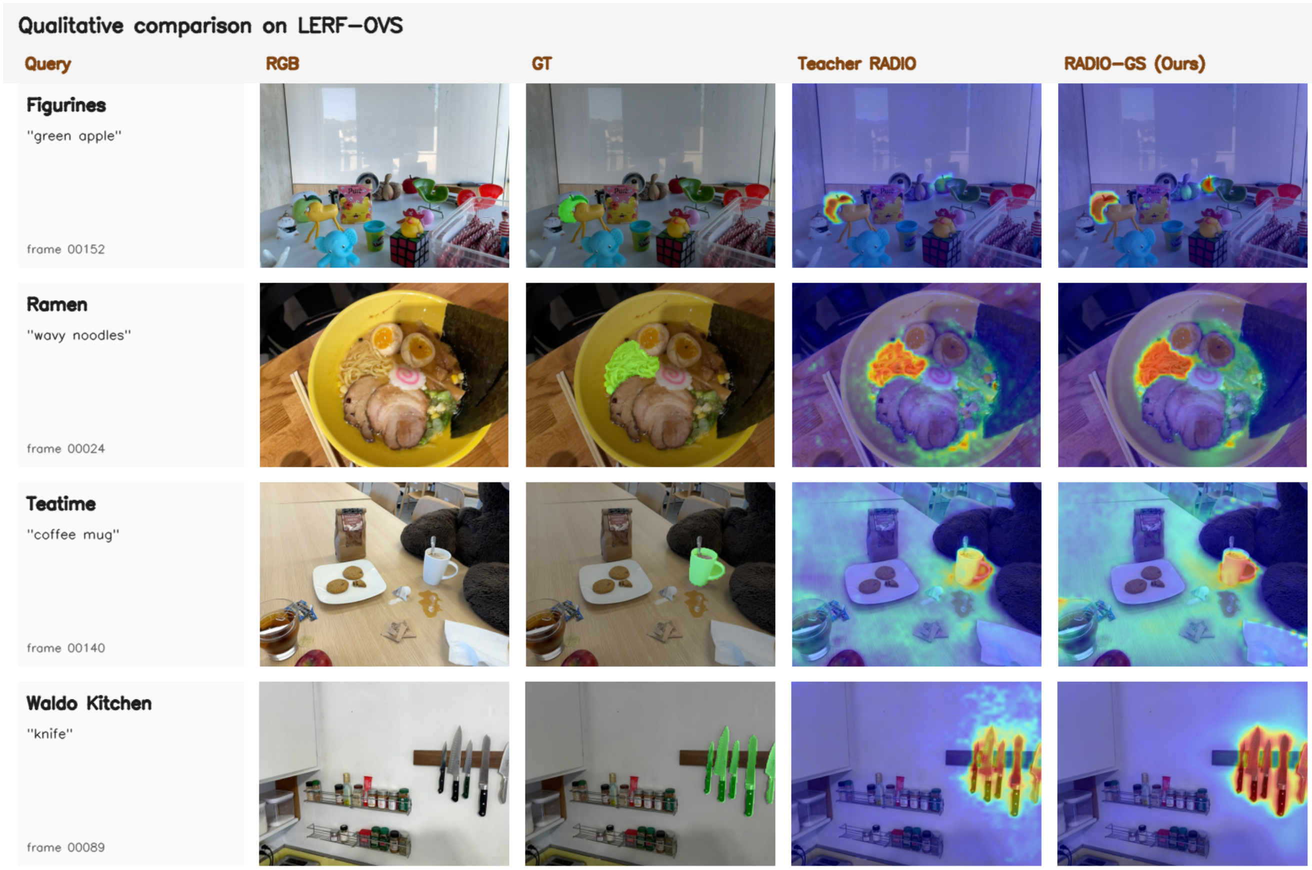

Figure 2. Qualitative open-vocabulary grounding examples from the calibrated LERF overlay runs. Each row shows the RGB image, ground-truth mask overlay, frozen C-RADIOv4-H teacher heatmap, and rendered CTF-GS heatmap for a representative query. Rendered masks use the same GT-free threshold-0.60 boundary calibration as Table 1.

5. Discussion

Why feature reconstruction matters. The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localization. The ScanNet results suggest that the same representation can be probed outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying once rendered features are registered back to Gaussians, but the current score-threshold selection is still weaker than explicit instance- or super-Gaussian grouping on fragmented scenes.

External-baseline scope. The LERF, LangSplat, and LEGaussians comparison rows are now traced to official paper or supplementary tables. They still come from cross-paper protocols, so the paper treats them as source-anchored context rows. A strict SOTA claim would require rerunning those baselines under the same evaluator and annotations used by RADIO-GS.

6. Limitations

CTF-GS depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because the teacher feature resolution and rendered feature alignment limit heatmap sharpness. The current ScanNet protocol is a direct point-query feature probe, not a replacement for a full standardized segmentation benchmark. The registered LERF direct 3D object-selection readout substantially improves primitive querying and slightly exceeds the OpenGaussian official macro reference under the reported OpenGaussian-style context, but it remains weaker on Waldo Kitchen and is not a locally rerun same-evaluator comparison. The paper should therefore avoid claiming universal primitive-level SOTA. Confidence-weighted VPR-to-field consistency improves the direct student field substantially, but it does not yet fully replace the streamed registered VPR readout on all scenes. Finally, the external comparison is official-source but not reproduced under one evaluator, so strong SOTA-style claims require additional baseline reruns. The stor-

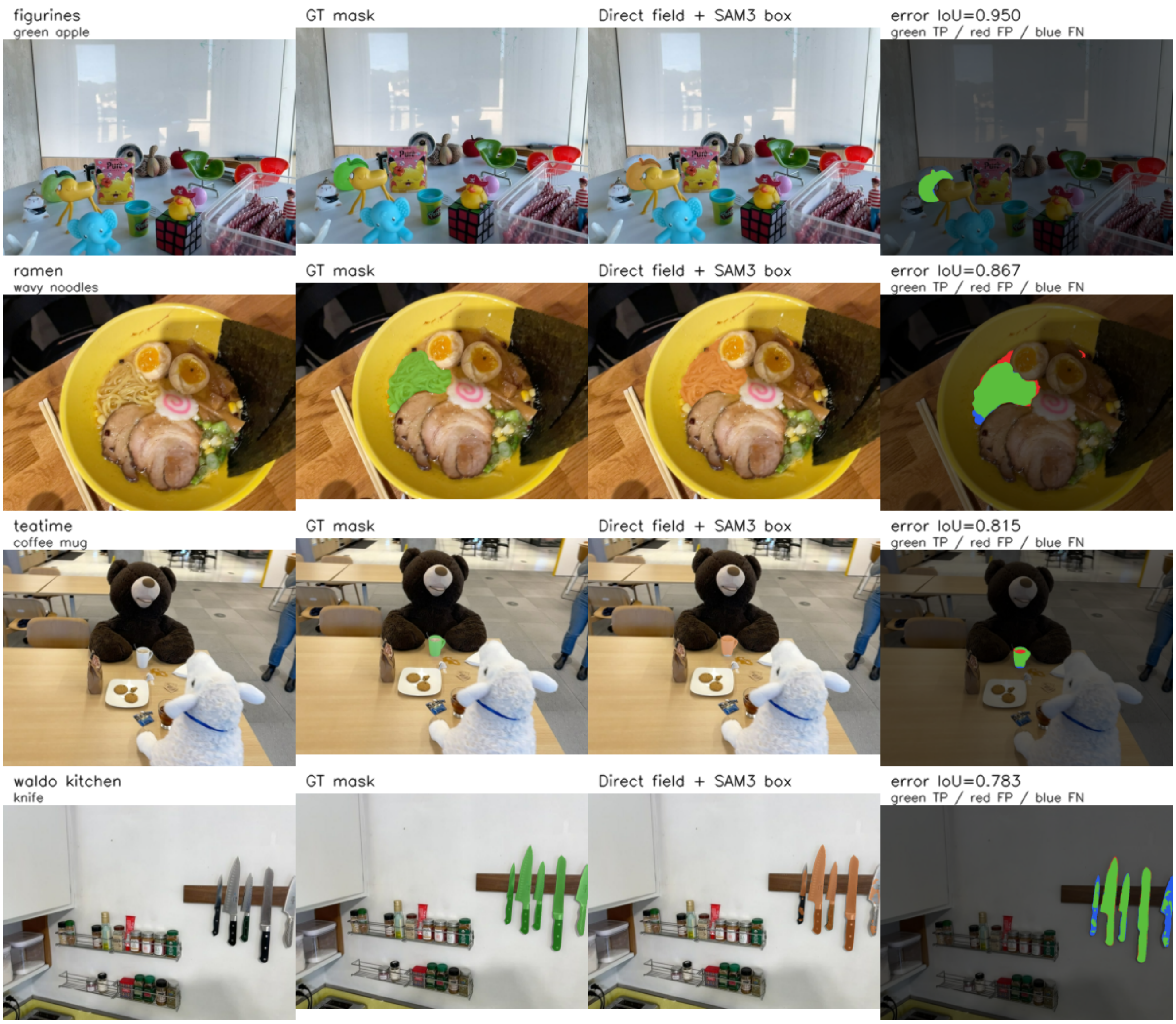

Figure 3. Qualitative LERF direct 3D object selection with compact direct-field scores and frozen official SAM3 box-prompt readout. Each row fixes a scene, annotated frame, and query, then shows RGB, ground-truth mask, the rendered direct-field selection refined by the SAM3 box readout, and an error map. Green is true positive, red is false positive, and blue is false negative. The Waldo Kitchen row illustrates that boundary quality improves but clutter and fragmentation remain the main failure mode.

age table reports checkpoint footprint rather than a deployed runtime memory allocator, so future packaging should separate feature-field parameters, rendering buffers, and optional downstream heads. The training entry point is now a modularized release artifact: the reproducibility audit verifies manifest, split, trusted-checkpoint, metrics-history, training-lock, and wrapped tensor-cache guards, and the entry script is reduced to 3735 lines with support code under `radio_gs/training`.

7. Conclusion

CTF-GS turns a 3D Gaussian scene into a reusable foundation-feature memory. By reconstructing frozen RADIO features rather than training a scene-specific classifier, it supports open-vocabulary grounding and cross-domain feature queries from novel views. The current freeze package provides a coherent submission path: strong LERF-OVS internal results, fair ScanNet direct-query evidence, explicit storage and runtime profiles, failure analysis, and source-anchored external comparison caveats.

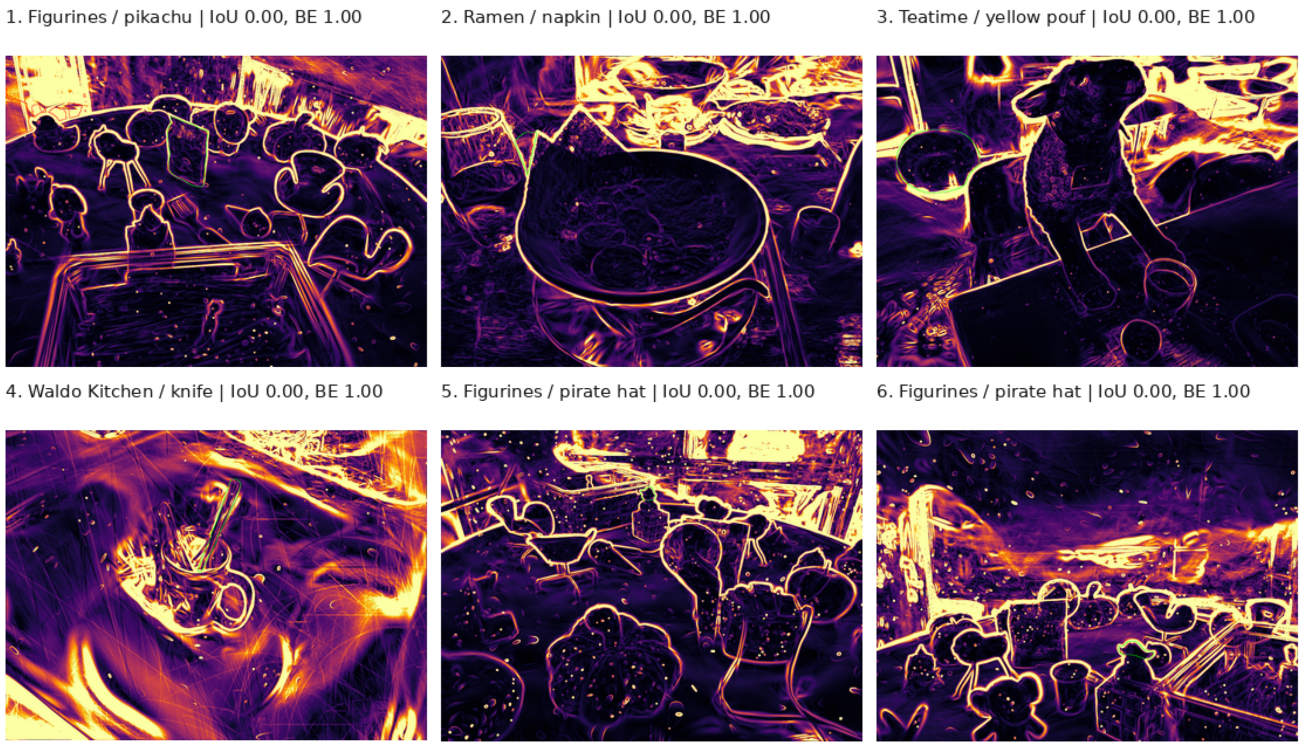

Figure 4. Selected alpha/depth discontinuity boundary cases for the strict pad16 SAM3-box direct-3D readout. Cases are selected from Table 12's worst boundary-error records, with scene coverage enforced before filling remaining slots. The overlays show that high discontinuity often co-occurs with failure cases, but the weak query-level correlations keep this figure as mechanism context rather than a causal proof.

References

[1] Jiazhong Cen, Jiemin Fang, Chen Yang, Lingxi Xie, Xiaopeng Zhang, Wei Shen, and Qi Tian. Segment any 3d gaussians. *arXiv preprint arXiv:2312.00860*, 2023. 2

[2] Yen-Jen Chiou, Wei-Tse Cheng, and Yuan-Fu Yang. Profuse: Efficient cross-view context fusion for open-vocabulary 3d gaussian splatting. *arXiv preprint arXiv:2601.04754*, 2026. 3

[3] Tianyu Huang, Runnan Chen, Dongting Hu, Fengming Huang, Mingming Gong, and Tongliang Liu. Openinsgaussian: Open-vocabulary instance gaussian segmentation with context-aware cross-view fusion. *arXiv preprint arXiv:2510.18253*, 2025. 2

[4] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 2023. 2

[5] Justin Kerr, Chung Min Kim, Ken Goldberg, Angjoo Kanazawa, and Matthew Tancik. Lerf: Language embedded radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. 2

[6] Jun-Seong Kim, GeonU Kim, Yu-Ji Kim, Yu-Chiang Frank Wang, Jaesung Choe, and Tae-Hyun Oh. Dr. splat: Directly referring 3d gaussian splatting via direct language embedding registration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2, 6

[7] Haijie Li, Yanmin Wu, Jiarui Meng, Qiankun Gao, Zhiyao Zhang, Ronggang Wang, and Jian Zhang. Instancegaussian: Appearance-semantic joint gaussian representation for 3d instance-level perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2, 6

[8] Kunyi Li, Michael Niemeyer, Sen Wang, Stefano Gasperini, Nassir Navab, and Federico Tombari. Opengaff: Open-vocabulary gaussian feature field with codebook attention. *arXiv preprint arXiv:2605.06088*, 2026. 6

[9] Siyun Liang, Sen Wang, Kunyi Li, Michael Niemeyer, Stefano Gasperini, Hendrik P. A. Lensch, Nassir Navab, and Federico Tombari. Supergseg: Open-vocabulary 3d segmentation with structured super-gaussians. *arXiv preprint arXiv:2412.10231*, 2026. 2

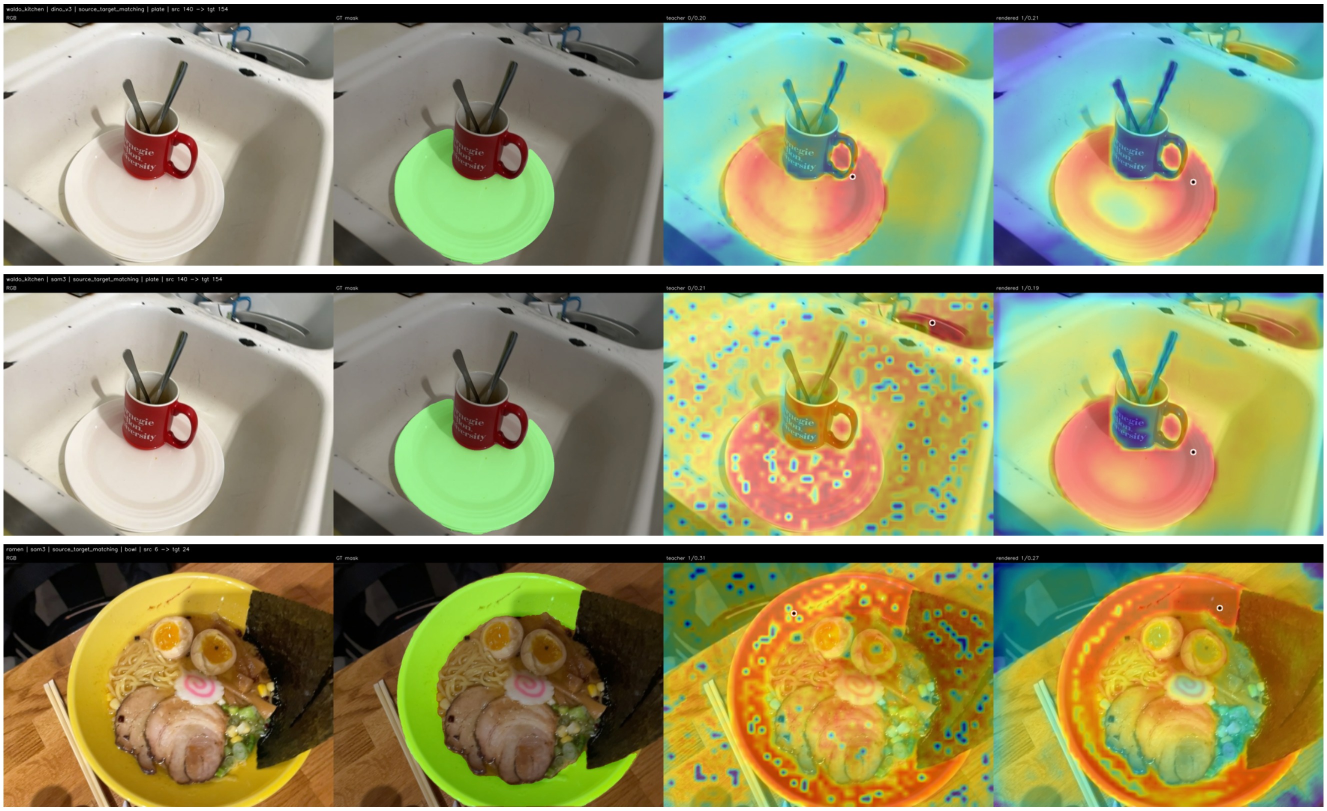

Figure 5. Qualitative DINOv3/SAM3 adaptor-space source-target matching. Each row shows RGB, GT mask, original RADIO teacher heatmap, and CTF-GS rendered heatmap. The Waldo Kitchen rows show cases where rendered features move the peak onto the target object; the Ramen row illustrates that the aggregate SAM3 gap remains visible even when localization is correct.

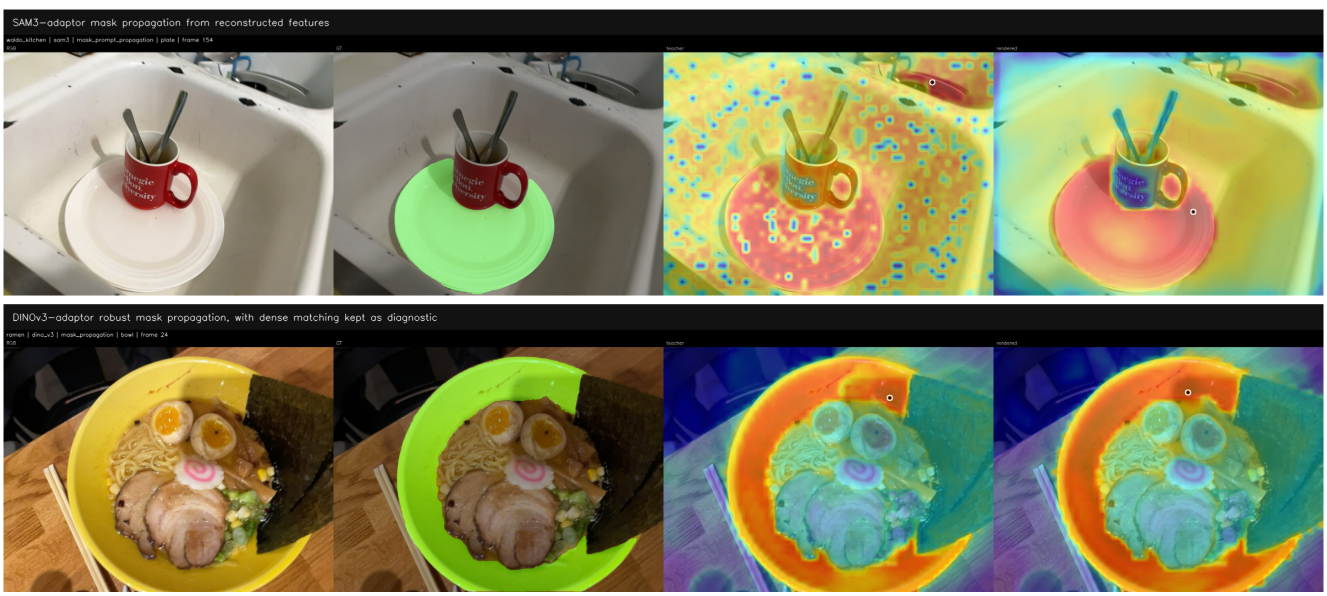

Figure 6. Qualitative promptable downstream probes from reconstructed features. Top: SAM3 mask-prompt propagation from a source annotation to a target view. Bottom: DINOv3 source-mask propagation with the promoted background-suppressed readout. Dense correspondences and homography-RANSAC filtering are kept as diagnostic visualizations rather than the main quantitative DINO evidence.

[10] Binh Long Nguyen, Kien Nguyen, Sridha Sridharan, Clinton Fookes, and Peyman Moghadam. Ilov3splat: Instance-level open-vocabulary 3d scene understanding in gaussian splatting. *arXiv preprint arXiv:2605.04506*, 2026. 2

[11] NVIDIA. C-radiov4: Tech report. *arXiv preprint arXiv:2601.17237*, 2026. 2

[12] Jens Piekenbrinck, Christian Schmidt, Alexander Hermans, Narunas Vaskevicius, Timm Linder, and Bastian Leibe. Opensplat3d: Open-vocabulary 3d instance segmentation using gaussian splatting. *arXiv preprint arXiv:2506.07697*, 2025. 2

[13] Minghan Qin, Wanhua Li, Jiawei Zhou, Haoqian Wang, and Hanspeter Pfister. Langsplat: 3d language gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 2

[14] Mike Ranzinger, Greg Heinrich, Jan Kautz, and Pavlo Molchanov. Am-radio: Agglomerative vision foundation model – reduce all domains into one. *arXiv preprint arXiv:2312.06709*, 2024. 2

[15] Jin-Chuan Shi, Miao Wang, Hao-Bin Duan, and Shao-Hua Guan. Language embedded 3d gaussians for open-vocabulary scene understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 2

[16] Yanmin Wu, Jiarui Meng, Haijie Li, Chenming Wu, Yahao Shi, Xinhua Cheng, Chen Zhao, Haocheng Feng, Errui Ding, Jingdong Wang, and Jian Zhang. Opengaussian: Towards point-level 3d gaussian-based open vocabulary understanding. *arXiv preprint arXiv:2406.02058*, 2024. 2, 6

[17] Xingxing Zuo, Pouya Samangouei, Yunwen Zhou, Yan Di, and Mingyang Li. Fmgs: Foundation model embedded 3d gaussian splatting for holistic 3d scene understanding. *arXiv preprint arXiv:2401.01970*, 2024. 2